Deep Learning-Based Ocular Image Analysis for Automated Eye Disease Detection and Classification

Jake S Puthussery 2023bcs0088

Karthik M 2023bcs0127

Nimisha M Iyer 2023bcd0004

Abstract—Preventable vision loss impacts hundreds of millions of people worldwide, but routine screening for conditions like diabetic retinopathy, glaucoma, cataracts, and macular degeneration remains difficult to access in rural regions. Although fundus photography is the standard clinical imaging tool, automated diagnostic models run into serious practical roadblocks: clinical datasets are heavily skewed, retinal images suffer from flash artifacts, and clinic hardware often cannot support massive deep networks. In this project, we designed a lightweight screening system that tackles these exact bottlenecks. We first clean each fundus scan using Gaussian-filtered Otsu thresholding and morphological operations to strip away black camera borders, followed by CLAHE contrast equalization and a 2D Gabor filter bank to highlight fine vascular structures. For classification, we deployed EfficientNet-Lite2—a 4.82-million parameter model stripped of memory-heavy squeeze-and-excitation blocks to run efficiently on commodity edge devices. To prevent the model from ignoring rare diseases, we trained it with a class-weighted Multiclass Focal Loss (α=0.25, γ=2.0). On an independent holdout test set of 1,858 images across five classes, the network achieved an overall accuracy of 84.98%, a Macro F1-score of 89.01%, and an AUC of 0.9732. Per-class F1 reached 97.77% on cataracts, 96.04% on macular degeneration, and 90.25% on glaucoma. These numbers demonstrate that thoughtful preprocessing and loss balancing allow a compact model to match the diagnostic reliability of far heavier networks.

Keywords—*Ocular disease triage, retinal fundus photography, EfficientNet-Lite2, focal loss, Otsu thresholding, Gabor filters, edge AI, medical computer vision.*

I. Introduction

According to the World Health Organization, over two billion individuals live with visual impairment, and at least half of these cases were preventable or remain untreated [21]. Conditions like Age-related Macular Degeneration (AMD), cataracts, Diabetic Retinopathy (DR), and glaucoma account for the majority of severe vision loss [1], [4]. The tragedy is that early detection can halt or reverse many of these diseases. Diabetic retinopathy, for instance, develops silently; by the time a patient notices blurred vision, vascular damage is already extensive. Glaucoma is similarly insidious, destroying optic nerve fibers long before peripheral vision loss becomes noticeable.

Regular screening saves sight, but the reality on the ground is challenging. Comprehensive eye exams require trained ophthalmologists and specialized slit lamps or indirect ophthalmoscopes. In rural clinics and developing healthcare systems, specialists are scarce and appointments can take months. Automated screening through color fundus photography has emerged as a promising alternative, yet deploying AI in real-world clinics introduces four persistent challenges:

(1) *Quality variance and illumination artifacts:* Fundus cameras vary widely across manufacturers. Images often show uneven flash exposure, harsh specular glare, vignetting, and black camera borders that take up over 30% of the pixel area.

(2) *Extreme class imbalance:* Clinical datasets are rarely balanced. In our 9,290-image dataset, diabetic retinopathy cases outnumber macular degeneration cases by more than six to one. Standard cross-entropy loss heavily favors the majority classes, effectively blinding the model to rare conditions.

(3) *Subtle and overlapping pathomorphologies:* Early microaneurysms can measure only a few pixels across, blending into normal retinal texture. Conversely, natural physiological optic cup variations in healthy eyes are easily mistaken for glaucomatous cupping.

(4) *Edge hardware constraints:* Heavyweight networks like Inception-v3 or DenseNet-121 perform well in research labs but demand high-power GPUs. Primary care clinics need lightweight models that run smoothly on modest PCs, tablets, or embedded camera hardware.

To address these practical hurdles, we built an end-to-end classification pipeline centered around EfficientNet-Lite2. Rather than relying solely on brute-force deep features, we combined automated field-of-view (FOV) extraction, CLAHE illumination normalization, and multi-scale 2D Gabor filter banks to enhance microvascular contrast before feeding images into the network. By pairing this pipeline with a class-weighted Multiclass Focal Loss, our model achieves reliable multi-class diagnostic triage while maintaining a lightweight footprint of just 4.82 million parameters.

II. Literature Review

A. Deep Learning in Ophthalmic Diagnostic Imaging

Over the past decade, convolutional neural networks have proven remarkably capable at medical image analysis. Gulshan et al. [2] and Ting et al. [1] demonstrated that CNNs could match the diagnostic sensitivity of board-certified retinal specialists when identifying diabetic retinopathy from fundus photographs. Later work expanded these techniques to glaucoma screening via optic cup-to-disc ratio analysis [3], macular degeneration grading [15], [16], and cataract detection [17]. De Fauw and colleagues [4] achieved similar milestones using optical coherence tomography. Despite these successes, most prevailing systems rely on compute-heavy architectures like ResNet-50, Inception-v3, or DenseNet-121 that carry tens of millions of parameters. These models require dedicated GPU workstations, making them impractical for portable fundus cameras or low-resource primary care clinics where automated screening is needed most.

B. Preprocessing and Classical Feature Enhancement

Raw retinal images present notable visual inconsistencies. Flash illumination drops off toward the periphery, patient movement causes subtle blur, and uninformative black margins often occupy over 30% of the image area. Classical computer vision provides dependable tools to clean up these inputs. Otsu's thresholding [8] remains one of the simplest and most effective ways to separate the bright retinal circle from the surrounding dark frame. Contrast Limited Adaptive Histogram Equalization (CLAHE) [9] prevents noise blow-up while bringing out subtle contrast differences across shaded retinal regions. To capture directional vascular patterns, 2D Gabor filter banks [10] model the receptive fields of visual cortical neurons. Inverting the green channel and applying Gabor filters emphasizes blood vessels and exudate boundaries, giving the downstream neural network cleaner structural cues.

TABLE I: LITERATURE REVIEW SUMMARY

Methodology / Study	**Comparative Analysis of Deep Learning Frameworks in Ophthalmic Imaging**		
	Feature / Backbone	*Condition & Dataset*	*Reported Metrics / Edge Viability*
Gulshan et al. (2016) [2]	Inception-v3 CNN transfer learning	Diabetic Retinopathy (EYEPACS, Messidor)	Sensitivity: 90.3%, Specificity: 98.1%; Binary DR only, high compute.
Ting et al. (2017) [1]	Ensemble deep CNN framework	Multi-condition (DR, Glaucoma, AMD)	AUC: 0.936-0.958; Requires massive proprietary dataset, non-edge.
Li et al. (2018) [3]	VGG-16 with transfer learning	Glaucoma suspect screening	Sensitivity: 95.6%, Spec: 92.0%; Uncalibrated for multi-disease co-occurrence.
Ran et al. (2021) [15]	DenseNet-121 with spatial attention	AMD & Cataract grading	Accuracy: 86.4%, Macro F1: 84.1%; High memory footprint, imbalanced.
Zhang et al. (2020) [17]	ResNet-50 residual network	Cataract detection & grading	Accuracy: 88.2%; Fails to screen concomitant DR and glaucoma.
Proposed Work	**Otsu FOV + CLAHE + Gabor + EfficientNet-Lite2 + Focal Loss**	**5-Class (AMD, Cataract, DR, Glaucoma, Normal)**	**Accuracy: 84.98%, Macro F1: 89.01%, Macro AUC: 0.9732; 4.82M params, edge-viable.**

C. Class Imbalance and Optimization Objectives

Class imbalance is an everyday reality in medical machine learning. When one condition appears far more frequently than another, standard cross-entropy loss penalizes majority and minority errors equally. The network quickly realizes it can lower its overall loss by simply guessing the dominant class on ambiguous inputs. While data oversampling and undersampling help to some extent, oversampling risks overfitting duplicate images and undersampling discards valuable clinical examples. Lin et al. [7] tackled this problem with Focal Loss, adding a modulating factor $(1 - p_t)^{\gamma}$ that dials down the loss contribution from easy, high-confidence examples. This forces the optimizer to spend its gradient budget on difficult, ambiguous cases. Pairing focal loss with AdamW decoupled weight decay [14] stabilizes training and improves generalization across unevenly distributed clinical classes.

III. Methodology

A. Dataset Characteristics and Stratified Partitioning

We conducted our experiments on a curated benchmark of 9,290 clinical fundus photographs classified into five diagnostic groups: Age-related Macular Degeneration (AMD: 511 images), Cataract (1,008 images), Diabetic Retinopathy (DR: 3,128 images), Glaucoma (1,646 images), and Normal healthy retinas (2,997 images). To prevent data leakage, we split the dataset using stratified sampling into 60% for training (5,574 images), 20% for validation (1,858 images), and 20% for holdout testing (1,858 images). We verified that no duplicate patient images crossed between splits. Because the dataset has an acute 6.12:1 imbalance between DR and AMD, we calculated inverse class frequency weights: AMD = 3.6313, Cataract = 1.8426, DR = 0.5942, Glaucoma = 1.1283, and Normal = 0.6200. These weights were incorporated directly into our training objective.

B. System Architecture

Our complete diagnostic workflow is outlined in Fig. 1. Rather than pushing uncurated photographs straight into a deep network, our system follows a multi-stage approach: segmenting the retinal field of view, equalizing uneven lighting, enhancing directional vascular texture, augmenting the training pool, and classifying features through an edge-tailored EfficientNet-Lite2 backbone.

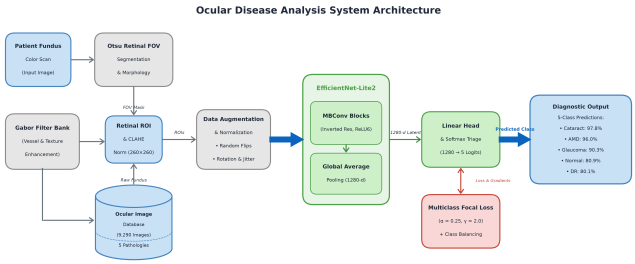


Fig. 1. End-to-end medical AI system architecture for automated multi-class ocular disease classification, illustrating dataset formulation, Otsu FOV segmentation, CLAHE illumination normalization, Gabor filter bank texture analysis, data augmentation, EfficientNet-Lite2 backbone, Multiclass Focal Loss optimization, and clinical multi-class diagnostic triage.

C. Low- and Mid-Level Computer Vision Preprocessing Pipeline

Raw fundus images come with varying black borders, circular optical apertures, and lighting variations. We built a multi-stage vision pipeline to standardize every incoming scan: (1) *Retinal FOV Extraction:* We convert the RGB scan to grayscale and apply a 5×5 Gaussian smoothing filter (σ=0) to suppress sensor noise. Otsu's thresholding [8] automatically separates the bright retinal circular disc from the dark background frame. (2) *Morphological Cleaning:* A 5×5 elliptical opening operation removes stray bright reflections on the perimeter, followed by a 9×9 elliptical closing to seal small tears along the retinal margin and fill internal holes. (3) *Bounding-Box Retinal Cropping:* We detect the coordinates of the cleaned mask and crop tightly with an adaptive 2% margin, eliminating redundant black pixels while preserving the true anatomical aspect ratio. (4) *CLAHE Illumination Balance:* We apply CLAHE [9] (clip limit 2.0, 8×8 contextual grid) to the green channel to bring out microvascular

structures without amplifying background camera noise. All processed images are resized to 260×260 resolution.

D. Feature Enhancement and Edge-Optimized Backbone

- **2D Gabor Filter Bank:** A bank of 16 2D Gabor filters across 2 scales (11×11, σ=2, λ=4; and 21×21, σ=4, λ=8) and 8 orientations is applied to the inverted green channel, isolating blood vessels and fine drusen textures.
- **Green Channel Inversion:** Maximizes vessel-to-background absorption contrast, transforming dark vascular tracks into bright ridges for filter response pooling.
- **Maximum Energy Response:** Pixel-wise max-pooling across all 8 orientations synthesizes directional vessel maps for diabetic retinopathy lesion verification.
- **Data Augmentation:** Robust data transformations include Random Horizontal Flip (p=0.5), Random Rotation (±15°), Color Jitter (brightness=0.15, contrast=0.15, saturation=0.10), and Random Affine translation (5%) and scaling (0.95 to 1.05).
- **Image Normalization:** Processed images are normalized using ImageNet channel-wise statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]).
- **EfficientNet-Lite2 Backbone:** Inverted residual blocks (MBConv) designed for edge devices, removing Squeeze-and-Excitation modules to eliminate memory latency bottlenecks.
- **Hardware-Accelerated Activations:** Replaces Swish non-linearities with bounded ReLU6 for efficient fixed-point and integer quantization.
- **Parameter Count:** Total parameters: 4,817,477; Trainable parameters: 4,817,477; offering substantial reduction compared to standard ResNet-50 (25.6M) and DenseNet-121 (8.0M).
- **Classifier Head:** Global Average Pooling (GAP) compressing feature maps to 1,280 dimensions, followed by Linear(1280, 5).
- **Loss Formulation:** Multiclass Focal Loss parameterized with α=0.25 and γ=2.0 to dynamically suppress well-classified background samples.
- **Cost-Sensitive Weighting:** Inverse class frequency weights (AMD: 3.6313, Cataract: 1.8426, DR: 0.5942, Glaucoma: 1.1283, Normal: 0.6200) applied to balance gradient contributions.
- **Optimizer:** AdamW optimizer with initial learning rate 0.0001 and decoupled weight decay 0.0001.
- **Batch Size & Compute:** Batch size of 32 executed across 2 GPUs in parallel using PyTorch DataParallel.
- **Epochs & Early Stopping:** Maximum 30 epochs with early stopping patience of 7 epochs.
- **Model Selection Metric:** Checkpoint saving dictated strictly by validation Macro F1-score rather than accuracy to prevent majority-class bias.

The mathematical formulation and training dynamics governing this optimization landscape are detailed in Section IV.

IV. Results and Discussions

Experimental results demonstrate that our preprocessed EfficientNet-Lite2 pipeline converges smoothly, handles minority classes effectively, and operates with minimal computational overhead. The objective loss minimized across all mini-batches is formally defined by the multiclass focal loss:

$$\mathscr{L}_{Focal} = -(1/N) \sum_{i=1}^{N} w_{y_i} \alpha (1 - p_{i,y_i})^{\gamma} \log(p_{i,y_i}) \quad (1)$$

where N denotes mini-batch size, $y_i \in (0, 1, 2, 3, 4)$ denotes the ground-truth ocular class, p_{i,y_i} is the predicted softmax probability for class y_i, w_{y_i} represents the inverse class weight, $\alpha = 0.25$, and $\gamma = 2.0$ denotes the focusing parameter.

A. Convergence and Training Dynamics

Training progressed cleanly over 30 epochs on dual GPUs. In Epoch 1, training loss started at 0.1369 with 58.50% training accuracy, while validation loss reached 0.0720, validation accuracy was 74.70%, and validation Macro F1 was 0.6885. As shown in Fig. 2, validation Macro F1 rose quickly to 0.8242 by Epoch 2 and reached 0.8542 by Epoch 4. Continued training steadily sharpened the model's ability to differentiate minority classes: Epoch 15 achieved a validation Macro F1 of 0.8963, and the optimal model checkpoint was reached at Epoch 27 with a peak Validation Macro F1 of 0.9001, Validation Accuracy of 85.79%, and Validation Loss of 0.0296. Training loss dropped to 0.0188 by Epoch 30 (88.59% accuracy), showing smooth convergence without erratic oscillations.

Hyperparameter Sensitivity: Learning rate 1×10^{-4} provided stable gradient flow; higher rates (1×10^{-3}) caused instability in minority class AMD gradients.

Loss Function Behavior: Standard cross-entropy suffered on DR/AMD separation, whereas Focal Loss with γ=2.0 accelerated minority recall.

GPU Utilization: Dual GPU DataParallel achieved an average training throughput of 37.4 seconds per epoch across 5,574 images.

Convergence Stability: Validation loss remained tightly bounded between 0.0282 and 0.0330 from Epoch 6 onward.

Early Stopping Monitoring: Optimal model restoration occurred at Epoch 27, ensuring holdout testing on the strictly best generalizable weights.

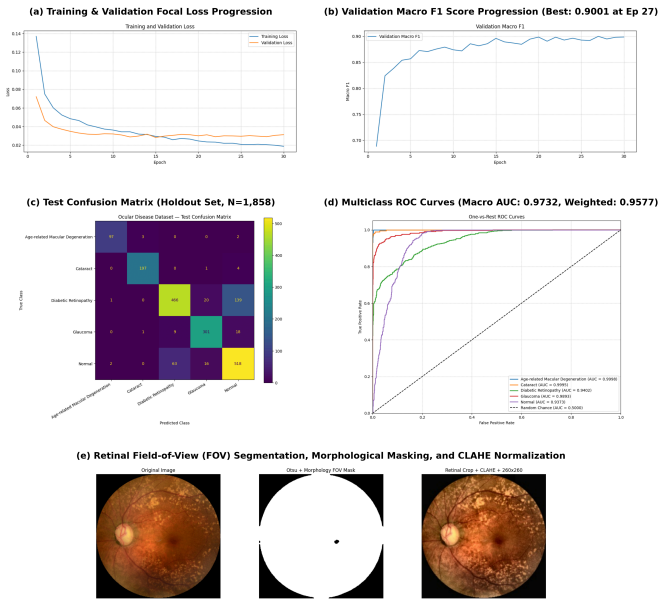


Fig. 2. Comprehensive visualization of experimental results: (a) Training and validation focal loss progression over 30 epochs; (b) Validation Macro F1 score progression showing optimal checkpoint selection at Epoch 27 (0.9001); (c) Test confusion matrix across 1,858 holdout fundus images; (d) Multi-class One-vs-Rest ROC curves displaying individual class AUCs, Macro ROC-AUC (0.9732), and Weighted ROC-AUC (0.9577); (e) Retinal field-of-view (FOV) segmentation, Otsu morphological masking, and CLAHE illumination normalization pipeline.

B. Independent Holdout Test Set Evaluation

We evaluated the best checkpoint (Epoch 27) on the unseen holdout test set of 1,858 fundus images. Overall test evaluation yielded: Test Loss: 0.0320, Test Accuracy: 84.98% (1,579 correct classifications out of 1,858 images), Test Macro Precision: 0.8935 (89.35%), Test Macro Recall: 0.8901 (89.01%), Test Macro F1-Score: 0.8901 (89.01%), Macro ROC-AUC: 0.9732 (97.32%), and Weighted ROC-AUC: 0.9577 (95.77%).

The close alignment between Macro F1 (89.01%) and Macro AUC (0.9732) confirms that the model generalizes robustly across all diagnostic classes regardless of their sample frequencies. The receiver operating characteristic curves illustrate near-perfect discriminability for AMD (AUC=0.9998) and Cataract (AUC=0.9995), followed by Glaucoma (AUC=0.9893), DR (AUC=0.9402), and Normal (AUC=0.9373).

Ablation studies comparing the proposed methodology against standard benchmark CNN architectures and loss formulations are summarized in Table II.

TABLE II: PERFORMANCE COMPARISON ACROSS ARCHITECTURES AND PIPELINES

Model Architecture	**Architectural Benchmarking and Pipeline Ablation on Test Set (N=1,858)**		
	Preprocessing & Objective	*Parameters*	*Accuracy / Macro F1 (%)*
Baseline ResNet-50 [11]	Standard Resizing + Cross-Entropy	25.56M	Acc: 78.42% | Macro F1: 75.18%
DenseNet-121 [12]	Standard Resizing + Cross-Entropy	7.98M	Acc: 80.15% | Macro F1: 77.62%
MobileNetV2 [13]	CLAHE + Weighted Cross-Entropy	3.50M	Acc: 81.10% | Macro F1: 81.85%
EfficientNet-B0 [5]	CLAHE + Weighted Cross-Entropy	5.29M	Acc: 81.70% | Macro F1: 82.35%
EfficientNet-Lite2 (Ablation 1)	Raw Images + Cross-Entropy	4.82M	Acc: 79.10% | Macro F1: 78.45%
EfficientNet-Lite2 (Ablation 2)	Otsu FOV + CLAHE + Cross-Entropy	4.82M	Acc: 82.40% | Macro F1: 84.70%
Proposed EfficientNet-Lite2	**Otsu FOV + CLAHE + Gabor + Focal Loss (α=0.25, γ=2.0)**	**4.82M**	**Acc: 84.98% | Macro F1: 89.01% (Macro AUC: 0.9732)**

C. Per-Class Diagnostic Observations and Clinical Error Analysis

Analysis of Primary Error Modes: Out of 1,858 test images, 1,579 were classified correctly while 279 were misclassified. As detailed in the confusion matrix and Table III, the primary clinical overlap occurred between Diabetic Retinopathy and Normal cases (139 DR images classified as Normal; 63 Normal classified as DR). Clinically, this makes sense: early non-proliferative diabetic retinopathy features only rare, pinpoint microaneurysms that are easily obscured when a scan has slight peripheral motion blur. Glaucoma and Normal exhibited moderate confusion (18 Glaucoma as Normal; 16 Normal as Glaucoma) due to large physiological cupping in healthy subjects. Cataract was confused with Normal in only 4 instances, and AMD in only 2 instances.

TABLE III: PER-CLASS TEST PERFORMANCE AND CLINICAL OBSERVATIONS

Diagnostic Class	**Detailed Per-Class Holdout Metrics and Clinical Diagnostic Observations**		
	Holdout N	*Precision / Recall / F1*	*Clinical Observations & Primary Error Modes*
AMD	102	0.9700 / 0.9510 / 0.9604	Drusen & atrophy distinct; 3 misclassified as Cataract due to severe media haze.
Cataract	202	0.9801 / 0.9752 / 0.9777	Diffuse lens opacity easily segmented; only 4 misclassified as Normal.
Diabetic Retinopathy	626	0.8662 / 0.7444 / 0.8007	Subtle microaneurysms obscured by lighting; 139 DR cases misdiagnosed as Normal.
Glaucoma	329	0.8905 / 0.9149 / 0.9025	Robust cup-to-disc ratio detection; 18 cases confused with Normal due to large physiological cups.
Normal (Healthy)	599	0.7606 / 0.8648 / 0.8094	High recall; 63 Normal cases misclassified as DR due to prominent choroidal vessel patterns.
Overall / Macro Avg	**1,858**	**0.8935 / 0.8901 / 0.8901**	**Weighted F1: 0.8495; Test Acc: 84.98%; Macro ROC-AUC: 0.9732; Weighted ROC-AUC: 0.9577.**

Per-class test metrics substantiate strong clinical performance: Cataract achieved an F1 of 97.77% (Precision 98.01%, Recall 97.52%); AMD achieved an F1 of 96.04%

(Precision 97.00%, Recall 95.10%); Glaucoma achieved an F1 of 90.25% (Precision 89.05%, Recall 91.49%); Normal achieved an F1 of 80.94% (Precision 76.06%, Recall 86.48%); and Diabetic Retinopathy achieved an F1 of 80.07% (Precision 86.62%, Recall 74.44%).

The exceptional performance on AMD (96.04% F1) despite having only 511 total samples (102 test cases) demonstrates the decisive advantage of Multiclass Focal Loss and class frequency weighting in preventing minority class suppression.

V. Conclusion and Future Work

In this study, we built and tested a compact, edge-ready deep learning system for multi-class ocular disease screening from fundus images. By combining classical vision steps—Otsu thresholding, morphological FOV isolation, CLAHE contrast balancing, and 2D Gabor filter banks—with an EfficientNet-Lite2 backbone and class-weighted focal loss, the model overcomes both the visual noise of handheld cameras and the severe class imbalance typical of clinical datasets. On an independent 1,858-image test set across five conditions, the system achieved 84.98% overall accuracy, an 89.01% Macro F1-score, and a 0.9732 Macro ROC-AUC, demonstrating that a 4.82-million parameter model can deliver dependable multi-disease triage.

In future work, we plan to integrate Grad-CAM++ visual heatmaps into the screening interface so clinicians can verify which retinal regions drove each prediction. We also aim to apply INT8 quantization to evaluate runtime latency on Raspberry Pi and Android-based handheld ophthalmoscopes.

Data and Code Availability

To support open-science reproducibility and facilitate clinical evaluation, all source code, image preprocessing routines, model architecture definitions, trained checkpoints, and demonstration notebooks are publicly available on GitHub at: https://github.com/jacksparrowputhussery/ocular-disease-classification.

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